



Curtin University

A photograph of a tea plantation. In the foreground, a young tea plant with vibrant green leaves grows from dark, rich brown soil. The plant is in sharp focus. In the background, rows of similar tea plants stretch across a valley, leading towards rolling hills under a bright blue sky with scattered white clouds. The lighting suggests a sunny day, with a warm glow on the right side of the image.

Curtin Credentials
**Introduction to Environment, Land,
Water Management and Closure
Planning over the Life of an Asset (LoA)**

Make tomorrow better.

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CRICOS Provider Code 00301J



Start:
Register your
Interest



Duration:
Approximately 30
hours



Delivery Mode:
Online



Price:
\$2269 (ext. GST)

Join Curtin University's introductory micro-credential in Environment, Land, and Water Management, designed for professionals building a foundation in mine closure planning.

Delivered by industry experts and facilitators, this micro-credential offers an introductory exploration of Environment, Land, and Water Management (ELWM) and Closure Planning within the context of an asset's entire life cycle (LoA). Comprising eight online modules, it equips you with a basic framework of the concepts in asset management while addressing environmental concerns, ensuring regulatory compliance, and optimising asset closure outcomes.

Across a broad spectrum of concepts and practical strategies, you'll discover how mining intersects with environmental responsibility and regulatory compliance, and how to manage stakeholder engagement. You'll explore the challenges and risks of asset management and build a holistic understanding of its complexities.

You'll gain practical insight into:

- The strategic and regulatory landscape of mine closure in Australia
- Asset levers and mine planning across the life of an asset
- Water management, landforms, and tailings
- Fundamental processes, risks, and controls
- Environmental impact assessments and site investigations
- Biodiversity, completion criteria, and rehabilitation trials

With a strong emphasis on an integrated, multidisciplinary approach, this credential forms part of the continuum of Technical Knowledge for Mine Closure Planning.

Learning Outcomes

Closure Planning

Describe the scope of Environment, Land, and Water Management (ELWM) and closure planning across the life of an asset, including the critical asset levers that determine closure outcomes and why planning must begin early in a mine's life.

Water, landforms and tailings

Explain approaches to water treatment and long-term monitoring of water resources through closure, and the modelling, hazard identification and reclamation of post-mining landforms.

Risk and impact assessment

Apply risk-based decision-making to closure considerations with reference to the ICMM Integrated Mine Closure guide, and develop a working knowledge of the environmental impact assessment process and its role in decision-making.

Biodiversity and rehabilitation

Develop the ability to apply international restoration standards to enhance biodiversity in post-mining landscapes, and evaluate how early rehabilitation trials contribute to favourable closure outcomes.

Who is this credential for?

This credential is suitable for:

- Established mining industry professionals
- Aspiring mining industry professionals
- Mining industry service providers (e.g. finance, investment, consultancies)
- First Nations peoples

Assessment

To successfully complete this credential, you are required to pass a final assessment.

To demonstrate what you have learned, you will:

- Develop a comprehensive report that critically analyses the integrated approach to Environment, Land, and Water Management (ELWM) and Closure Planning across the Life of an Asset (LoA).

Earn a badge

Once you successfully complete and pass the final assessment, you'll earn a digital badge that is instantly shareable to your social networks (including LinkedIn) which showcases your new skills and knowledge mastery.



Extend

This credential provides foundation knowledge in a discipline and doesn't require previous knowledge.

Discover more

Curtin Credentials focus on five themes, which have been carefully curated based on what's most relevant and valuable to professionals today, and in the future. This credential fits within the following theme:



Future of Work

Thrive and adapt in the changing world of work by developing and broadening your skillset in a range of areas including work design

Meet your facilitators



Dr Michael Just

Dr Michael Just is a restoration ecologist at Curtin University, researching seed biology, plant propagation and tissue culture to address the technical bottlenecks limiting ecological restoration in Western Australia. He works with mining, infrastructure and government partners to develop propagation pathways for species that are ecologically critical but difficult to produce at scale, and has led restoration planning and species-supply strategy for major offset projects, including multi-decade rehabilitation programs governed by conservation agreements.



Dr George Barakos

Dr Barakos has significant experience not only in the academia but also in the mining industry, having worked as a consulting engineer in mining operations in Greece for several years. To keep pace with the evolution in the mining industry and its transition to the digital era, his recent research interest is also focusing on the automation and digitalisation of mining operations, as well as the digitalisation of the technical, economic and environmental evaluation of mining projects; how to introduce the IoT in mining operations, digitalisation of the mining project evaluation process and commodity market analyses using big data.



Zane Hughes

Zane is a Waanyi Traditional Custodian of North West Queensland and has held leadership roles in a range of mining companies and infrastructure project development companies and in Indigenous corporations representing Native Title holders. He has an in-depth understanding and knowledge of the mining sector, processes, and operational drivers. Aside from his role as a Project Lead Researcher at the Cooperative Research Centre for Transformations in Mining Economies (CRC TiME), Zane has served on a number of committees and represented Native Title interests across a range of national forums and is also a Research Fellow for the Faculty of Science and Engineering at Curtin University.

Meet your facilitators



Dr Apurna Ghosh

Dr Apurna Ghosh has taught and researched in the mining sector for more than two decades. At Curtin he teaches Mine Safety Management, Mine Planning and Design, Resource Estimation and Mining Systems. His research identifies workplace risk factors across different mining settings, including dust control and its effects on miners and their families, and he has published over 50 international technical articles on mine safety. He also works in geoinformatics, including digital elevation modelling for aquifer recharge site selection and catchment monitoring.



Chris Rawson

Chris Rawson is an environmental scientist whose career bridges research and innovative education. After completing a PhD in remediated site assessment at the University of Technology Sydney, he joined Curtin University in 2009 as a postdoctoral researcher, investigating the impacts of marine pollution, from dredging to oil and gas developments, on ecologically and commercially important fish populations on Australia's Northwest Shelf. Since 2015 he has held senior leadership positions in learning and teaching at Curtin, including Associate Dean (Learning and Teaching) for the School of Science and Dean of Learning and Teaching for the Faculty of Science and Engineering.



Dr. Mohammadali Sepehri

Dr Mohammadali (Ali) Sepehri is a Lecturer at Curtin University and a Professional Mining Engineer (P.Eng). He holds MSc and PhD degrees in Mining Engineering from the University of Alberta, Canada, focused on advanced geomechanical numerical modelling. Ali has worked on numerous geomechanical mining projects, including open pit and underground mine stability design, ground control and support design, geomechanical monitoring and instrumentation, and prediction of mining-induced surface subsidence. His research interests include rock engineering design, slope stability, the geotechnical aspects of mine tailings, underground thermal energy storage and numerical modelling.





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